

Group Theoretic Analysis of Degeneracy and Splitting in the Nuclear Shell Model

The Nuclear Shell Model provides a framework for understanding the arrangement of nucleons into discrete energy levels. A rigorous way to analyze these levels; specifically their degeneracy and the subsequent energy splitting is through the lens of group theory. By examining the evolution of the nuclear potential from a simple oscillator to a realistic well with spin-orbit interactions, we can track how fundamental symmetries are preserved or broken.

1 The Isotropic Harmonic Oscillator

As an initial approximation, we model the nucleus using an Isotropic Harmonic Oscillator. The Hamiltonian for this system is given by:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 r^2$$

The Hamiltonian is invariant not only under spatial rotation, but under the much larger group $SU(3)$. We can express this by decoupling the dimensions:

$$H = \sum_{i=1}^3 \left(\frac{p_i^2}{2m} + \frac{1}{2}m\omega^2 x_i^2 \right) = H_x + H_y + H_z$$

Since the dimensions decouple, the total energy is simply the sum of these three decoupled systems:

$$E_N = \left(N + \frac{3}{2} \right) \hbar\omega \quad ; \quad N = n_x + n_y + n_z$$

Here, energy ONLY depends on N . This is the reason the Hamiltonian belongs to $SU(3)$ symmetry.

To find the spatial states, for some fixed value of n_x take K ; where $0 \leq K \leq N$. This leaves $n_y + n_z = N - K$, yielding one more possibility since zero is included.

$$\text{Total spatial states} = \sum_{K=0}^N (N - K + 1) = \frac{(N+1)(N+2)}{2}$$

Now, the question is how do the (n_x, n_y, n_z) states map to the spherical (n, l, m) parity states. In Cartesian coordinates, Parity $= (-1)^N$, while in spherical coordinates, Parity $= (-1)^l$. This implies l must have the same parity as N . Therefore, for a given N , the allowed states of orbital angular momentum drop by 2:

$$l = N, N-2, N-4, \dots \quad (\text{down to } 0 \text{ or } 1)$$

2 The Infinite Square Well

To represent the sharp boundary of a nucleus more accurately, we transition to an Infinite Square Well. By changing the potential from $V(r) \propto r^2$ to $V(r) = 0$ for $r < R$ and ∞ for $r \geq R$, we destroy the symmetry of the Hamiltonian.

Since the Hamiltonian no longer decouples, the freedom of allowed states with different angular momentum to share the same energy is gone. However, an infinite square well is still a perfect sphere. Hence, it retains $SO(3)$ symmetry, where the degeneracy is:

$$2(2l + 1)$$

The reason the energy levels drop is that the harmonic oscillator (HO) has a parabolic bowl of a potential, whereas inside the well the potential now is perfectly flat. Therefore, the average energy (potential) of a nucleon in this well is strictly lower than it would've been in the HO well.

Looking at the effective potential, $V_{\text{centrifugal}} \propto \frac{l(l+1)}{r^2}$. Thus, the higher the l , the higher the V_{eff} , creating extra repulsion.

- **In a Harmonic oscillator:** The 1f state has a very strong centrifugal push outwards towards the boundary which forces it to climb up the steep, quadratic wall $V(r) \propto r^2$. This costs a lot of potential energy (P.E). For the 2p state, the centrifugal push isn't much. Hence it sits much closer to the centre; however, due to the presence of a node, it costs kinetic energy (K.E). These two balance each other out to become degenerate.
- **In an infinite well:** The 1f state, which was initially forced to climb up the quadratic wall, since the ground is now flat, doesn't cost any energy to stay at the boundary, drastically reducing the energy requirement. Flattening the ground doesn't change the surroundings of the 2p state drastically, giving no significant drop.

Due to this symmetry breaking, group theory dictates that the originally large $SU(3)$ representation must branch into a direct sum of smaller $SO(3)$ representations.

3 The Finite Square Well and Boundary Conditions

A finite square well continues to preserve the $SO(3)$ symmetry, meaning l remains a 'good' quantum number.

However, the boundary conditions have changed. Initially, finding a particle at the boundary had zero probability since no particle can have infinite energy. Now, at the boundary, the probability density is exponentially decaying but not zero. The wave function hence spreads more than before.

4 The Woods-Saxon Potential

Further refining the model to match actual nuclear density profiles brings us to the Woods-Saxon potential. Even here there is no symmetry breaking, hence no energy splitting occurs.

A finite square well is a rectangle which has sharp 90° corners at the bottom and top. When the Woods-Saxon potential rounds off those corners, this makes the potential well wider at higher energies than it was before. Since now the wavefunction has more room to spread, the push of the centrifugal potential has more effect to push further outwards. From simulation, one can see for sharp edges at the boundary, the effective potential is much higher for higher l states. i.e., the more the rounding, the lower the effective potential, resulting in a lesser energy cost to be at the boundary. Hence, the centrifugal potential can, with lesser energy, push the higher l states towards the boundary.

5 Introducing Spin-Orbit Coupling

The final crucial step to reproduce experimentally observed nuclear states is spin-orbit coupling. Before we introduce it, the Hamiltonian of the nucleus in the Woods-Saxon potential is H_0 .

As of now, the orbital angular momentum $\vec{L}$ and spin angular momentum $\vec{S}$ live independently. Hence, the total symmetry group is a direct product of the two independent symmetry groups:

$$G_{\text{initial}} = \text{SO}(3)_L \otimes \text{SU}(2)_S$$

$$|n, l, m_l\rangle \otimes |s, m_s\rangle = |n, l, m_l, s, m_s\rangle$$

The total degeneracy for $s = \frac{1}{2}$ is given by $(2l+1) \times (2s+1)$, making the total $d = 2(2l+1)$. We now introduce spin-orbit coupling: $H = H_0 + V_{ls}(r)\vec{L} \cdot \vec{S}$. We can see that $[\vec{L}, H] \neq 0$ and $[\vec{S}, H] \neq 0$. However, the total angular momentum commutes: $[\vec{J}, H] = 0$, where $\vec{J} = \vec{L} + \vec{S}$.

Hence, the symmetry has collapsed into a smaller symmetry group: $\text{SU}(2)$. Now, we have the representation:

$$D^{(l)} \otimes D^{(1/2)} = D^{(l+1/2)} \oplus D^{(l-1/2)}$$

Consequently, the unperturbed state with orbital angular momentum l splits into two distinct states. To work in these new states, one has to go under a basis transformation:

$$|n, l, s, j, m_j\rangle = \sum_{m_l, m_s} C_{l, m_l, s, m_s}^{j, m_j} |n, l, m_l\rangle |s, m_s\rangle$$

This shifts the energy levels based on the expectation value:

$$\langle \vec{L} \cdot \vec{S} \rangle = \frac{\hbar^2}{2} [j(j+1) - l(l+1) - s(s+1)]$$

Producing a final energy gap of:

$$\Delta E_{\text{gap}} = \frac{\hbar^2}{2} \langle V_{ls}(r) \rangle (2l+1)$$